

Evidence Holonomy and Entropy Production: From Universal Coding to Irreversibility

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Abstract

We define an “evidence holonomy” functional on loops of representation transforms applied to sample paths, in two versions. **Representation-space holonomy** scores each representation with its own universal code; by pointwise universality it converges to the entropy-rate difference $h(Q) - h(P)$, where Q is the law of the loop’s output. **KL holonomy** instead scores the original path under the loop’s pushforward law; its rate is the relative-entropy rate $d(P||Q)$, and for the time-reversal loop of a stationary finite-state Markov chain it equals the entropy production rate σ (bits/step). Estimating KL holonomy requires a model of Q that is accurate on paths typical for P , which universality alone does not provide. We analyze the order- R plug-in estimator implemented in our code. It is consistent for σ on Markov chains for every $R \geq 1$. On records observed through a memoryless channel, averages of its order-wise limits are lower bounds on the hidden entropy production. Numerical checks on random chains, closed-form rings, reversible chains, and a hidden ring show convergence, a small positive finite-sample bias, and the gap between observed and hidden irreversibility.

1 Setup and Definitions

Alphabet and path space. Fix a finite alphabet $\mathcal{X}$. Let $\mathcal{X}^n$ denote length- n strings and $\mathcal{X}^{\mathbb{N}}$ the one-sided sequence space with its product σ -algebra. Let P be a stationary ergodic probability measure on $\mathcal{X}^{\mathbb{N}}$. Write $X_{0:n-1}$ for the length- n prefix of a sample from P and P_n for its law on $\mathcal{X}^n$.

Universal codes. A *universal code* on alphabet $\mathcal{A}$ is a map $\mathcal{E}_{\mathcal{A}} : \bigcup_{n \geq 1} \mathcal{A}^n \rightarrow \mathbb{R}_+$ assigning a code length in bits to any finite string, such that for every stationary ergodic law Q on $\mathcal{A}^{\mathbb{N}}$,

$$\frac{1}{n} \left(\mathcal{E}_{\mathcal{A}}(Y_{0:n-1}) + \log_2 Q_n(Y_{0:n-1}) \right) \xrightarrow[n \rightarrow \infty]{} 0 \quad Q\text{-a.s. and in } L^1(Q). \quad (1)$$

Classical examples include LZ78, Krichevsky–Trofimov (KT) mixtures over Markov orders, and context-tree weighting [1, 2, 3, 5, 6]. Note that (1) is a statement about paths typical for Q only.

Standing assumptions. Unless stated otherwise, alphabets are finite; processes are stationary and ergodic; and when relative-entropy rates are finite we assume $P_n \ll Q_n$ for all n . All logs are base 2, and all rates are per symbol.

Representation transforms and loops. For each n , let $F_{i,n} : \mathcal{X}_{i-1}^{n_{i-1}(n)} \rightarrow \mathcal{X}_i^{n_i(n)}$ be measurable, where the alphabets $\mathcal{X}_i$ may differ by step and $n_0(n) = n$. A finite list $\gamma = (F_{1,n}, \dots, F_{m,n})$ is a *loop at scale n* if $\mathcal{X}_m = \mathcal{X}_0 = \mathcal{X}$ and $n_m(n) = n + O(1)$. Write $x^{(i)} = F_{i,n} \circ \dots \circ F_{1,n}(x)$ and $L_n = F_{m,n} \circ \dots \circ F_{1,n}$. When $n_m(n) \neq n$, the two arguments being compared are aligned by dropping $O(1)$ boundary symbols, which does not affect per-symbol limits.

Why “holonomy”? In differential geometry, holonomy measures the failure of a vector to return unchanged after parallel transport around a closed loop. Here the “vector” is the description length (or log-likelihood) an observer assigns to a path, and the holonomy is its net change after the path is carried around a loop of representations. Bijective relabelings are “flat” (zero holonomy); time reversal of a nonequilibrium process is not. The analogy is motivational: we use it to organize two different ways of comparing evidence before and after a loop, not to claim a connection or curvature in the geometric sense.

1.1 Two holonomy functionals

Definition 1.1 (Representation-space holonomy). Given universal codes $\mathcal{E}_{\mathcal{X}_i}$ for the intermediate alphabets,

$$\text{Hol}_n^{\text{out},\gamma}(x) = \sum_{i=1}^m \left(\mathcal{E}_{\mathcal{X}_i}(x^{(i)}) - \mathcal{E}_{\mathcal{X}_{i-1}}(x^{(i-1)}) \right) = \mathcal{E}_{\mathcal{X}}(L_n(x)) - \mathcal{E}_{\mathcal{X}}(x).$$

Definition 1.2 (KL holonomy). Let $Q_n = (L_n)\#P_n$ be the law of the loop’s output. The (ideal) KL holonomy is

$$\text{Hol}_n^{\text{KL},\gamma}(x) = \log_2 \frac{P_n(x)}{Q_n(x)},$$

the code length of x under Q_n minus its code length under P_n . Its expectation is $\mathbb{E}_P[\text{Hol}_n^{\text{KL},\gamma}] = D(P_n \| Q_n)$.

Definition 1.2 involves the unknown laws P_n and Q_n ; Section 4 discusses how to estimate it from data.

2 Reductions

Lemma 2.1 (Representation-space holonomy). *Suppose the loop’s outputs are the n -marginals of a stationary ergodic process Q (see Section 3.2), and $\mathcal{E}_{\mathcal{X}}$ is universal. Then*

$$\frac{1}{n} \left(\text{Hol}_n^{\text{out},\gamma}(X_{0:n-1}) - \log_2 \frac{P_n(X_{0:n-1})}{Q_n(L_n(X_{0:n-1}))} \right) \xrightarrow{n \rightarrow \infty} 0 \quad \text{in } L^1(P). \quad (2)$$

If, in addition, the outputs $L_n(X_{0:n-1})$ are prefixes of a single Q -distributed sequence (as for sliding-block codes), the convergence also holds P -a.s.

Proof. Since $L_n(X_{0:n-1}) \sim Q_n$, the $L^1(Q)$ part of (1) applied to Q controls $\mathcal{E}_{\mathcal{X}}(L_n(X)) + \log_2 Q_n(L_n(X))$, and applied to P controls $\mathcal{E}_{\mathcal{X}}(X) + \log_2 P_n(X)$; subtract [4, 5]. L^1 convergence depends only on the marginal laws. Almost-sure convergence needs the outputs for different n to lie on one sample path. This fails for time reversal, where $L_n(x)$ is a different string for every n . $\square$

Theorem 2.2 (Expectation-level identities). *Under the assumptions of Lemma 2.1,*

$$\frac{1}{n} \mathbb{E}_P [\text{Hol}_n^{\text{out}, \gamma}] = h(Q) - h(P) + o(1). \quad (3)$$

For KL holonomy, whenever the limit exists,

$$\frac{1}{n} \mathbb{E}_P [\text{Hol}_n^{\text{KL}, \gamma}] = \frac{1}{n} D(P_n \| Q_n) \xrightarrow{n \rightarrow \infty} d(P \| Q) \geq 0. \quad (4)$$

Proof. Take expectations in (2), using $\mathbb{E}_P[-\log_2 P_n(X)] = H(P_n)$ and $\mathbb{E}_P[-\log_2 Q_n(L_n(X))] = H(Q_n)$, and divide by n . The second identity is Definition 1.2. $\square$

Remark 2.3 (Scope). Equation (3) measures net compression or expansion under the loop and does not depend on which universal code is used. Equation (4) is the *irreversibility* functional: it compares two laws on the same coordinates and is non-negative.

3 Canonical Loops

3.1 Gauge invariance for bijective loops

Corollary 3.1 (Gauge invariance). *If the loop is the identity map on $\mathcal{X}^n$ (for example, a relabeling followed by its inverse), then $Q_n = P_n$, so $\text{Hol}_n^{\text{KL}, \gamma} \equiv 0$ and $\frac{1}{n} \text{Hol}_n^{\text{out}, \gamma}(X_{0:n-1}) = 0$.*

3.2 Coarse-graining and observation channels

A deterministic coarse-graining followed by a lift, $L_n = R_n \circ K_n$, produces outputs supported on the image of R_n . Unless P_n is supported there too, $P_n \not\ll Q_n$ and the KL rate is infinite. So deterministic coarse-grain–lift loops do not give informative KL holonomies. (Estimators with smoothed probabilities return finite numbers that grow as the smoothing vanishes.)

The useful statement about coarse-graining concerns applying the *same* channel to both laws. Let C be a memoryless channel (a stochastic matrix applied independently to each symbol), and write CP for the law of the channel output when the input has law P . By the data-processing inequality [7],

$$D((CP)_n \| (CQ)_n) \leq D(P_n \| Q_n) \quad \text{for every } n. \quad (5)$$

When is the pushforward stationary? For a general sequence of maps L_n , the laws Q_n need not be the n -marginals of any stationary process. A sufficient condition is that the loop acts as a shift-commuting sliding-block code $(L_n(x))_t = F(x_{t-m:t+m})$. Time reversal is not a sliding-block code, but the reversed process of a stationary process is stationary, so it is covered directly.

3.3 Time reversal and entropy production for Markov chains

Let P be a stationary Markov chain on $\mathcal{X} = \{1, \dots, k\}$ with transition matrix T and stationary law π . Its time reversal P^{rev} is the Markov chain with transitions $T_{ij}^* = \pi_j T_{ji} / \pi_i$; equivalently, $P_n^{\text{rev}}(x_0, \dots, x_{n-1}) = P_n(x_{n-1}, \dots, x_0)$.

Time-reversal loop. The loop is literal reversal, $L_n(x_{0:n-1}) = (x_{n-1}, \dots, x_0)$, so $(L_n)\#P_n = P_n^{\text{rev}}$. It can also be written as a loop through the transition alphabet $\mathcal{X}^2$. Encode x as its edge sequence $e_t = (x_{t-1}, x_t)$, $t = 1, \dots, n-1$; reverse the edge sequence; then keep the *second* state of each edge. This yields $(x_{n-1}, x_{n-2}, \dots, x_1)$, the reversal of $x_{1:n-1}$, which differs from literal reversal by one boundary symbol. Our implementation uses this form and compares it against $x_{1:n-1}$. The edge presentation adds no information beyond literal reversal.

Theorem 3.2 (KL holonomy of time reversal equals entropy production). *Suppose $T_{ij} > 0 \Leftrightarrow T_{ji} > 0$ for all i, j . Then*

$$D(P_n \| P_n^{\text{rev}}) = (n-1)\sigma, \quad \sigma = \sum_{i,j} \pi_i T_{ij} \log_2 \frac{\pi_i T_{ij}}{\pi_j T_{ji}} \quad (\text{bits/step}), \quad (6)$$

so $d(P \| P^{\text{rev}}) = \sigma$ and $\frac{1}{n} \mathbb{E}_P[\text{Hol}_n^{\text{KL,rev}}] \rightarrow \sigma$.

Proof. For a path $x = x_{0:n-1}$ with positive probability,

$$\log_2 \frac{P_n(x)}{P_n^{\text{rev}}(x)} = \log_2 \frac{P_n(x_{0:n-1})}{P_n(x_{n-1}, \dots, x_0)} = \log_2 \frac{\pi_{x_0}}{\pi_{x_{n-1}}} + \sum_{t=1}^{n-1} \log_2 \frac{T_{x_{t-1}x_t}}{T_{x_t x_{t-1}}}.$$

Under stationarity, the boundary term has expectation zero and each summand has expectation $\sum_{i,j} \pi_i T_{ij} \log_2(T_{ij}/T_{ji})$. This equals σ because $\sum_{i,j} \pi_i T_{ij} \log_2(\pi_i/\pi_j) = \sum_i \pi_i \log_2 \pi_i - \sum_j \pi_j \log_2 \pi_j = 0$. This is the standard identification of entropy production with the relative entropy between forward and reversed path measures [8, 9, 10, 11]. $\square$

Remark 3.3 (Absolute continuity). If $T_{ij} > 0$ but $T_{ji} = 0$ for some pair, then $P_n \not\ll P_n^{\text{rev}}$ and the rate is $+\infty$.

Remark 3.4 (Why not $h(Q) - h(P)$?). Reversal preserves entropy rates, $h(P) = h(P^{\text{rev}})$, so representation-space holonomy vanishes for the time-reversal loop. Only the KL version detects irreversibility.

General stationary processes. For any stationary P with $P_n \ll P_n^{\text{rev}}$, the same definitions give $\frac{1}{n} \mathbb{E}_P[\text{Hol}_n^{\text{KL,rev}}] \rightarrow d(P \| P^{\text{rev}})$ whenever this limit exists.

4 Estimation

Universality is not enough. A natural idea is to estimate $\log_2 Q_n(x)$ with a universal code trained on samples from Q . But (1) controls code lengths only on paths typical for Q . When $P \neq Q$ are both stationary ergodic, they are mutually singular, and a P -typical path is Q -atypical. So universality says nothing about the code length a Q -trained coder assigns to $X \sim P$. Code-length *differences* from separate compressors (for example, LZ78 on x and on $L_n(x)$) estimate $h(Q) - h(P)$, which is zero for reversal. Estimating $d(P \| Q)$ requires conditional probabilities of Q that are accurate on contexts that occur under P .

The plug-in estimator. Given $x_{0:n-1}$, let $y = x_{1:n-1}$ and $\tilde{y} = (x_{n-1}, \dots, x_1)$ its reversal. For $r = 0, \dots, R$, let $\hat{p}_r(a | c)$ be the KT estimate $(N_y(c, a) + \frac{1}{2}) / (N_y(c) + \frac{k}{2})$ from the counts of context $c \in \mathcal{X}^r$ followed by a in y , and $\hat{q}_r$ the same from $\tilde{y}$. Mix orders with the Bayesian posterior weights

obtained by sequentially coding y (respectively $\tilde{y}$), then freeze the resulting predictors $\hat{p}$ and $\hat{q}$. The estimate is

$$\hat{\sigma}_{n,R} = \frac{1}{n-1} \sum_t \log_2 \frac{\hat{p}(y_t \mid y_{t-R:t-1})}{\hat{q}(y_t \mid y_{t-R:t-1})}, \quad (7)$$

where the first few symbols use shorter contexts. Both predictors are evaluated on the same y used to fit $\hat{p}$ (an in-sample plug-in estimate, cf. [13]).

Proposition 4.1 (Consistency for Markov chains). *Let P be a stationary irreducible Markov chain with $T_{ij} > 0 \Leftrightarrow T_{ji} > 0$, and let $R \geq 1$. Then $\hat{\sigma}_{n,R} \rightarrow \sigma$ almost surely.*

Proof sketch. By the ergodic theorem, the empirical frequencies of $(r+1)$ -blocks in y converge a.s. to those of P . Those in $\tilde{y}$ are the same blocks reversed, so they converge to those of P^{rev} . Both P and P^{rev} are first-order Markov. So for every $r \geq 1$ and every context of positive probability, $\hat{p}_r(\cdot \mid c) \rightarrow T_{c,r,\cdot}$ and $\hat{q}_r(\cdot \mid c) \rightarrow T_{c,r,\cdot}^*$, and the KT smoothing vanishes. If T is not rank one, the order-0 component's log-likelihood deficit grows linearly in n , so its posterior weight tends to zero. If T is rank one, the chain is i.i.d., $\sigma = 0$, and every component converges to π . Hence $\hat{p} \rightarrow T$ and $\hat{q} \rightarrow T^*$ uniformly over the finitely many contexts visited. By the support condition, $T_{ij}^* > 0$ whenever $T_{ij} > 0$, so the log-ratios stay bounded. The ergodic theorem applied to (7) gives the limit $\sum_{i,j} \pi_i T_{ij} \log_2(T_{ij}/T_{ij}^*) = \sigma$. $\square$

Remark 4.2 (Robustness to R , not coder independence). Every order $r \geq 1$ is consistent for a first-order chain, and the posterior concentrates on the smallest adequate order. So estimates for different $R \geq 1$ nearly coincide at moderate n ; Section 5 reports agreement to within 10^{-13} bits/step. This reflects a shared, correctly specified model class. It is not evidence that arbitrary universal codes yield the same KL estimate.

Proposition 4.3 (Observed records bound hidden entropy production). *Let Z be a stationary Markov chain satisfying the conditions of Theorem 3.2, with entropy production σ_Z , and let $Y = CZ$ be observed through a memoryless channel C . Write $D_m = D(P_m^Y \parallel P_m^{Y,\text{rev}})$. Then $D_m \leq (m-1)\sigma_Z$ for every m , so $d(P^Y \parallel P^{Y,\text{rev}}) \leq \sigma_Z$ whenever the rate exists.*

Proof. Because the channel acts independently on each symbol, reversing the observed path is the same as observing the reversed hidden path: $P_m^{Y,\text{rev}} = (CP^{Z,\text{rev}})_m$. Apply (5) and Theorem 3.2. $\square$

For an observed process Y , Proposition 4.1's argument shows that the order- R estimator converges (under ergodicity and positivity) to the conditional quantity $d_R = D_{R+1} - D_R$. Because $D_1 = 0$ (reversal preserves one-symbol marginals), the averages satisfy

$$\frac{1}{m-1} \sum_{R=1}^{m-1} d_R = \frac{D_m}{m-1} \leq \sigma_Z.$$

Individual terms d_R need not be monotone in R or below σ_Z . So a single finite-order estimate is not a certified bound, and the in-sample bias below pushes it upward.

Remark 4.4 (Finite-sample behavior). Evaluating $\hat{p}$ in-sample favors the forward model, so $\hat{\sigma}_{n,R}$ has a positive bias. A heuristic χ^2 argument suggests it is of order $k^{R'+1}/n$, with R' the order the mixture selects; for reversible chains ($\sigma = 0$), Section 5 shows it shrinking roughly in proportion to $1/n$. Fluctuations are of order $n^{-1/2}$. The code provides a block bootstrap for intervals.

5 Numerical validation

All numbers below come from the ancillary files in `anc/. scripts/paper_experiments.py` regenerates them with fixed seeds; the estimator uses $R = 3$ unless stated otherwise.

Random chains. Table 1 and Figure 1 follow 15 random biased chains (five each with 3, 4, and 5 states; σ between 0.0067 and 1.92 bits/step) as the sample length grows from 2^9 to 2^{15} . The median relative error falls from 10.3% at $n = 2^9$ to 1.3% at $n = 2^{15}$. Chains with small σ converge slowest in relative terms; the largest error at $n = 2^{15}$ is 19%.

Table 1: Absolute relative error $|\hat{\sigma} - \sigma|/\sigma$ of the time-reversal estimate ($R = 3$) on 15 random biased chains (five each with 3, 4, and 5 states), by sample length n . Every row uses the same 15 chains.

n	median	interquartile range	max
2^9	10.3%	[2.2%, 14.1%]	123%
2^{11}	8.8%	[5.3%, 11.8%]	86%
2^{13}	4.4%	[3.7%, 7.7%]	31%
2^{15}	1.3%	[0.3%, 3.9%]	19%

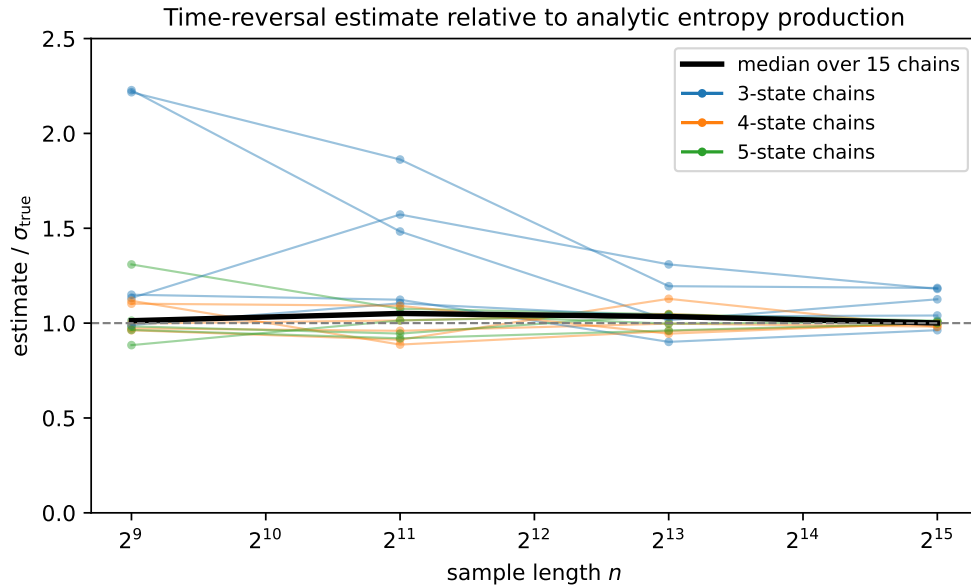


Figure 1: Estimate divided by the analytic σ for 15 random chains (thin lines, colored by state count) and their median (black), as the sample length grows.

Calibration. Figure 2 compares the estimate with the analytic σ for 60 further random chains (3–5 states, bias parameter drawn uniformly from $[0, 0.9]$) at $n = 2^{15}$. The correlation is $r = 0.9994$, and the median relative error is 1.9%. Relative errors are large only where σ itself is tiny: the worst case is a chain with $\sigma \approx 10^{-4}$ bits/step, whose absolute error is at the level of the in-sample bias below. For the 53 chains with $\sigma \geq 0.05$ bits/step, the median relative error is 1.7% and the largest is 10.4%.

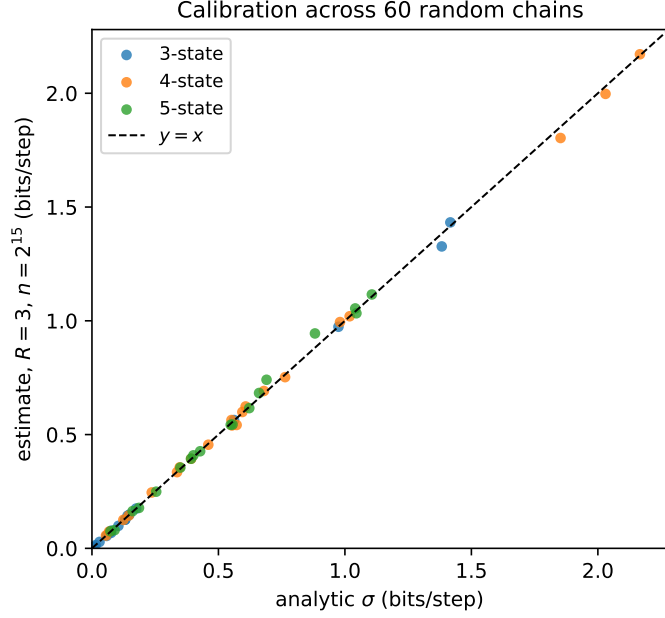


Figure 2: Estimate ($R = 3$, $n = 2^{15}$) against analytic entropy production for 60 random chains.

Reversible chains and bias. For 10 random detailed-balance chains of each size ($\sigma = 0$), the mean estimate with $R = 3$ is 0.0015 bits/step at $n = 2^{11}$ and 0.00013 at $n = 2^{15}$ for 3-state chains, and 0.0100 and 0.00072 for 5-state chains. Estimates with $R = 1$ and $R = 3$ are identical. The bias shrinks by a factor of 12–14 over a 16-fold increase in n , consistent with the roughly $1/n$ behavior described in Section 4.

Order robustness and closed forms. On first-order chains, estimates with $R = 1, 2, 3, 5$ agree to within 10^{-13} bits/step, as the remark in Section 4 predicts. The validation battery (`uec-battery --run-suite`) also checks:

- three-state rings with closed-form σ , to within 5% relative error at $n = 1.5 \times 10^5$;
- a reversible two-state chain (estimate below 5×10^{-4});
- gauge loops (below 5×10^{-3});
- agreement with a smoothed transition-count estimator;
- block-bootstrap coverage of σ ;
- negative controls: alphabet mismatch, empty outputs, and self-evidence differences, which stay near zero under reversal even when $\sigma > 0$.

Hidden-state records. Figure 3 observes a three-state ring ($p = 0.3$ forward, $q = 0.05$ backward, $\sigma_Z = 0.646$ bits/step) through two channels, for $n = 2^{17}$ and three seeds. Through a noisy channel that reports the true state with probability 0.8, the estimates for $R = 1, \dots, 6$ range from 0.085 to 0.127 bits/step, well below σ_Z . They rise from $R = 1$ to $R = 3$ and then stay the same: at this sample size the mixture puts negligible weight on orders above 3, so larger R adds nothing. Through a channel that merges states 1 and 2, the observation is symmetric under the relabeling

that maps the ring to its reversal. The observed process is therefore exactly reversible, and the estimates stay at most 7×10^{-5} bits/step even though $\sigma_Z > 0$.

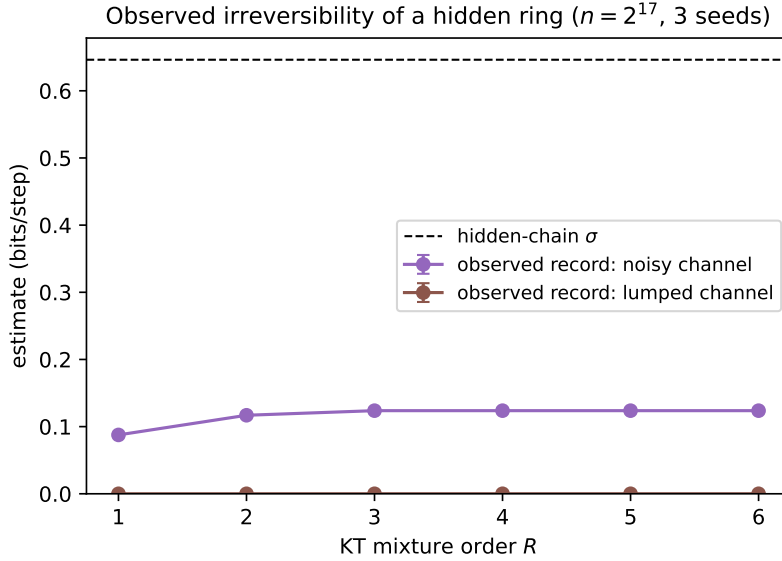


Figure 3: Estimates on observed records of a hidden three-state ring, by KT mixture order R (mean and range over three seeds), against the hidden chain’s entropy production.

Arrow-of-time demonstrations. The package also applies the estimator to quantized real-valued signals (audio, sensor, and price series; images and video via vector quantization). It compares held-out windows with their literal reversals and reports an AUC with a block-bootstrap interval. On i.i.d. noise with 10^4 samples, single-run AUCs ranged from 0.11 to 0.74 across seeds. So these demonstrations are illustrations whose intervals must be read, not validation of the theory.

6 Code and data

Code, ancillary data, and scripts are at <https://github.com/jbwinters/Evidence-Holonomy>. `scripts/reproduce_all.sh` regenerates `anc/`, the table, the figures, and this document.

Discussion

The two holonomies answer different questions. Evaluating each representation with its own universal code gives the entropy-rate difference $h(Q) - h(P)$: it is insensitive to the choice of code and detects net compression or expansion, but it vanishes for time reversal. Evaluating the original path under the loop’s pushforward law gives the relative-entropy rate $d(P\|Q)$, which for Markov time reversal is the entropy production rate. The second quantity is the one that captures irreversibility. It requires a consistent model of the reversed process on forward-typical contexts, and a finite-order plug-in estimator supplies one for finite-state Markov chains.

Extensions. Quantizing a continuous process and observing a Markov chain through a memoryless channel are both instances of (5). The observed irreversibility rate therefore bounds the

underlying entropy production from below [12]; the quantum analogue follows from the monotonicity of quantum relative entropy [14, 15]. Turning finite-sample, finite-order estimates into certified bounds would require controlling both the in-sample bias and the order-wise terms d_R .

Limitations. The results assume finite alphabets, stationarity, and ergodicity. The consistency result covers finite-state Markov chains; for hidden-state records, the estimator targets the conditional quantities d_R , not σ_Z . Estimates carry a positive finite-sample bias (empirically of order $1/n$) and $O(n^{-1/2})$ fluctuations. The loop formalism organizes these estimators but does not by itself supply new estimators for loops other than time reversal.

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